

# Purple E-Commerce Order Management System

Requirement Analysis and Customer Database Module

Student Name: FARJANA S

Registration Number: TU6253210211012

## 1. Introduction

The purpose of this document is to analyze and specify the business requirements of the Purple E-Commerce Order Management System. The system manages customers, sellers, products, orders, payments, and deliveries while ensuring secure and efficient order processing.

## 2. Project Scope

The system allows customers to register, browse products, add items to a cart, place orders, make secure payments, track deliveries, manage returns and refunds, and maintain complete order history.

## 3. Intended Users

- Customers
- Sellers
- Warehouse Staff
- Delivery Partners
- Customer Support Executives
- Administrators

## 4. Business Requirements

- Secure customer registration and login
- Maintain customer profiles and multiple delivery addresses
- Manage product catalog and inventory
- Process online payments
- Track orders in real time
- Support returns and refunds
- Generate sales and inventory reports
- Send order notifications via email/SMS

## 5. Functional Requirements

- FR-1 Register customers
- FR-2 Customer login
- FR-3 Search products
- FR-4 Shopping cart management
- FR-5 Place orders
- FR-6 Secure payment processing
- FR-7 Order tracking (Ordered → Packed → Shipped → Out for Delivery → Delivered)
- FR-8 Return and refund requests
- FR-9 View order history
- FR-10 Admin sales reports

## 6. Non-Functional Requirements

- Response time under 2 seconds
- Encrypted passwords and secure payment information
- 99.9% system availability
- Scalable for millions of customers
- Daily backup and disaster recovery

## 7. Customer Database Module

Fields:

customer\_id (PK), full\_name, email, phone\_number, password (encrypted), address, city, state, pincode, prime\_member, registration\_date

## 8. Real-Time Purple E-Commerce Scenario

Farjana logs into Purple, searches for a product, adds it to her cart, pays using UPI, and places the order. Purple records her details, updates inventory, generates an invoice, and lets her track the order through Ordered, Packed, Shipped, Out for Delivery, and Delivered. If needed, she can request a return or refund.

## **9. Assumptions and Constraints**

- Only registered customers can place orders.
- Every order belongs to one customer.
- Products must be in stock before purchase.
- Passwords are stored in encrypted form.

## **10. Conclusion**

This document defines the Week 1 requirements for the Purple E-Commerce Order Management System with a focus on the Customer Database Module. It provides the foundation for designing the ER diagram, database schema, and future modules such as Products, Orders, Payments, Inventory, Delivery, and Returns.